

Urgent repairs fixed on time out bush, without letting cost quietly decide who waits

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Remote repairs cost more, so they quietly fall to the back

70

remote communities in our model, with 4,481 public houses

8.5×

what an emergency repair costs in a very remote community, against a remote one (Nous Group, 2017)

96%

of an emergency job's cost can be travel alone (Nous Group, 2017)

6 h

drive from Katherine to Kalkarindji, one of the 16 communities its tradespeople serve

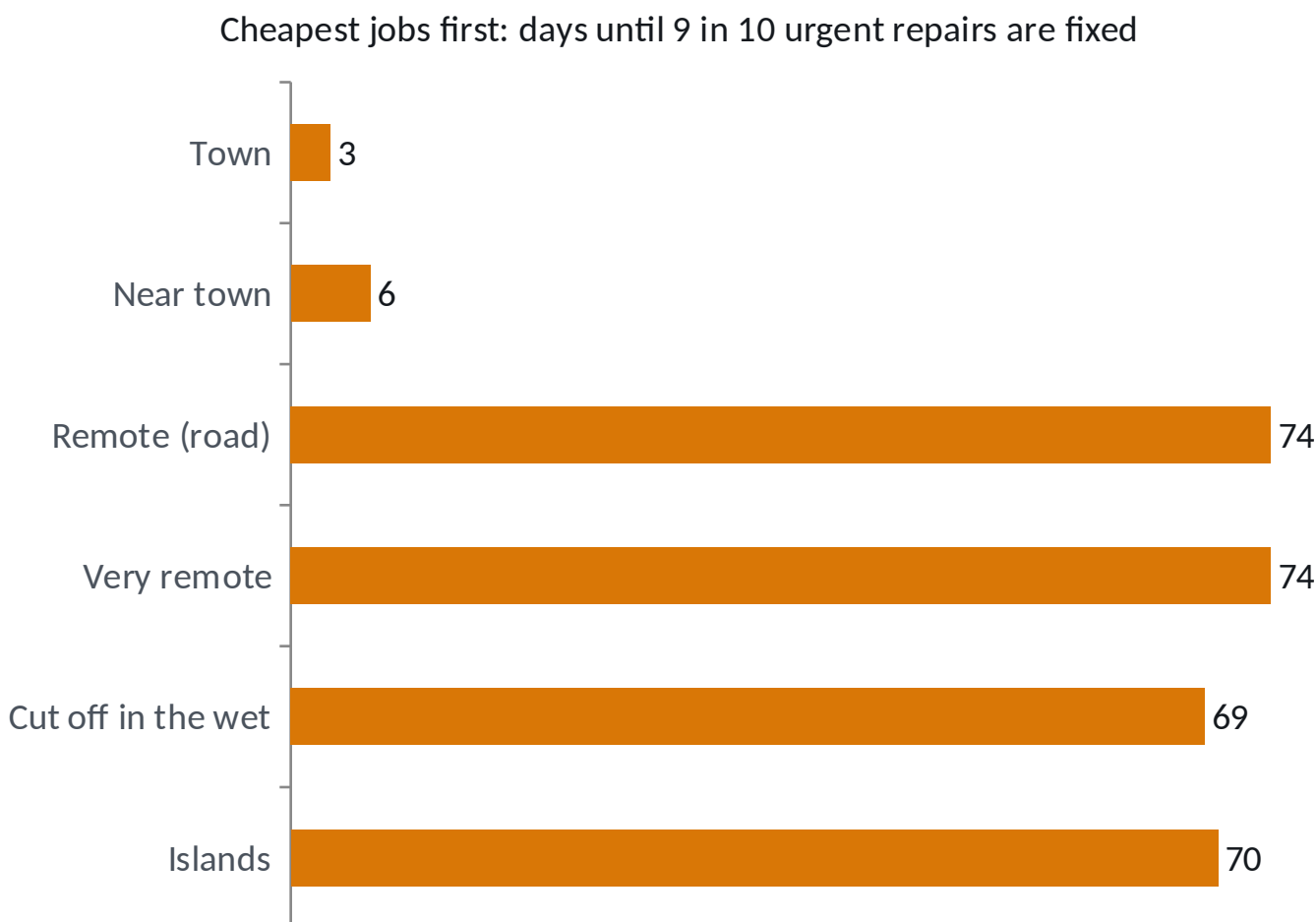


A real kind of message, from our simulated year

“Your no power to the house repair is urgent. Waiting 56 days. Held up by travel cost.”

For 28 days no electrician was sent to your community. It costs about \$8,100 to fly one there and back. That was a cost decision, not a judgement about your repair. Nobody signed off on it.

Plan for the cheapest jobs, and remote tenants wait weeks



62 days

for 9 in 10 urgent remote repairs, against 3 days in town

Why: a schedule built to fix the most jobs for the money always picks town first. It's efficient on paper, and no one ever decides that remote families should wait.

ReachNT: rank by need, share trips, and put a name on the trade-off



Read

Reads what the tenant said, in their words, and picks the fault and how urgent it is. Unsure? A person calls back.



Rank by need

Danger, health, who lives there and time waited set the order. Distance and cost are never in the score.



Plan shared trips

Each week it plans who goes where. Neighbouring communities share one trip, found with Uber's H3 hexagons.



Explain and sign

A named coordinator signs how much cost may count. Every tenant can see why their repair waited.

One tool, four views: the coordinator plans, the tradesperson gets a run sheet that works without signal, the tenant gets a straight answer, and the housing officer records for tenants without the app.

AI that reads tenants' words, and knows when to ask a person

What the tenant said

"no sparks now but the switch is black and kids touch it"

Word rules + a learning model

Rules catch known phrases; a logistic regression model catches wording the rules miss

Sure, and not dangerous?

Ranked straight away

Unsure or risky?

A person checks today. It never downgrades danger on its own

99.3%

of dangerous reports caught, on wording the AI had never seen, because doubtful ones go to a person

0.84

ROC-AUC for the model alone on new wording (1.00 on familiar)

42%

of unfamiliar reports sent to a person

We never show tenants a confidence percentage: when the model said it was 90% sure, it was right about half the time. Being honest about that is part of the design.

Fair to people who say less, or say it differently

This weekTripsNew reportChecksCompare

water coming through the...

toilet blocked pls come

sewage coming up in the ...

no hot water for the bab...

smells like gas inside t...

front door kicked in won...

cockroaches everywhere i...

pls come

Ask every time, the same way

Is anyone in danger right now?
(sparks, smoke, gas, water near power, sewage inside, something falling)

YesNoNot asked

Does anyone need power, cooling or medical supplies to stay well? (dialysis, insulin, oxygen, medicine kept in the fridge)

YesNoNot asked

Is there a baby under 12 months, or a frail elder? (an elder old enough for aged care)

YesNoNot asked

Are there young children, or is anyone pregnant, sick, or finding it hard to get around? (wheelchair, walking frame)

YesNoNot asked

How many people are living there now? (the tenancy record fills this in; ask if it may have changed)

10

Have you told anyone about this fault before? (the house's job history fills this in)

YesNoNot asked

Tenancy record for house 12: 10 people, 3 bedrooms. Change the number above if it has changed.

Told in full

"toilet blocked, my nana lives here, 9 of us, third time I rang"

50 → 5

points a household lost for saying less (average)

Told in a few words, in Kriol, to the housing officer

"toilet blocked pls come". Same house, same need. From the words alone: up to 45 points less

95% → 6%

of short reports ranked below the same household told in full

So everyone is asked the same six questions

Anyone in danger now? Needs power, cooling or medicine; a baby under 1 or a frail elder (Tier 1)? Young children, pregnant, sick or hard to get around (Tier 2)? How many live there? Told anyone before? Records fill in the rest. "Not asked" never costs points.

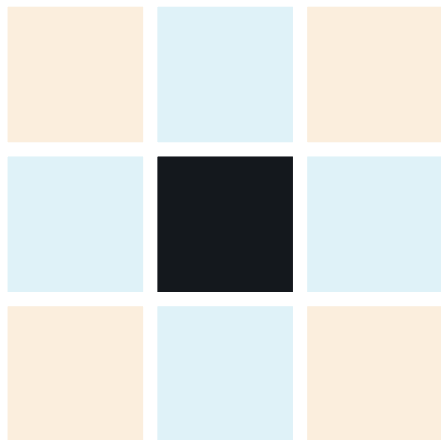
13% → 89%

of Tier 1 or 2 households recognised from a few words

Clock starts at first contact • free interpreters (AIS, TIS National, NRS) • how, when and in what language you report is never scored

Why we built it on Uber's H3 hexagon grid

Squares



8 neighbours at 2 different distances

Hexagons



6 neighbours, all the same distance away

Why it matters: “two hexagons away” is the same distance in every direction, so deciding which communities can share a trip is fair to all of them.



Shared trips

Communities within 2 hexagons (about 90 km) can share a trip, saving \$67–\$76 per repair



Privacy

A house is stored as a small hexagon, not a street address



Public numbers

Waits are published per larger hexagon, only where there are at least 5 repairs

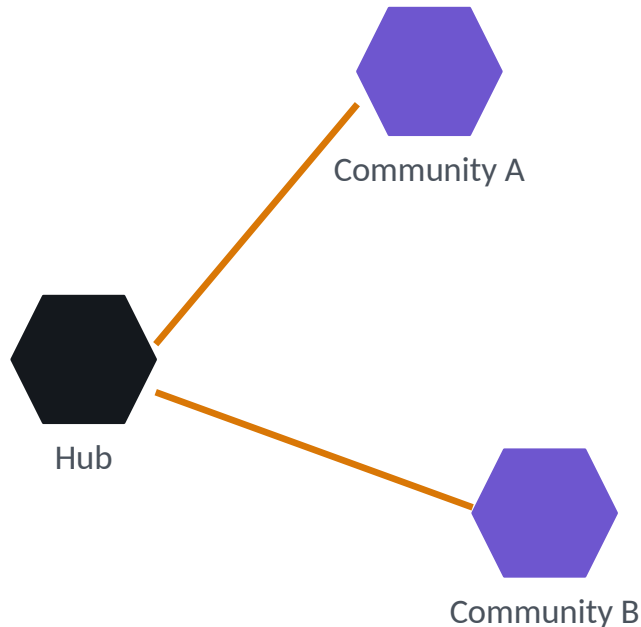


Field and maps

"Nearby jobs" for tradespeople, the driving-time map, and fast database lookups

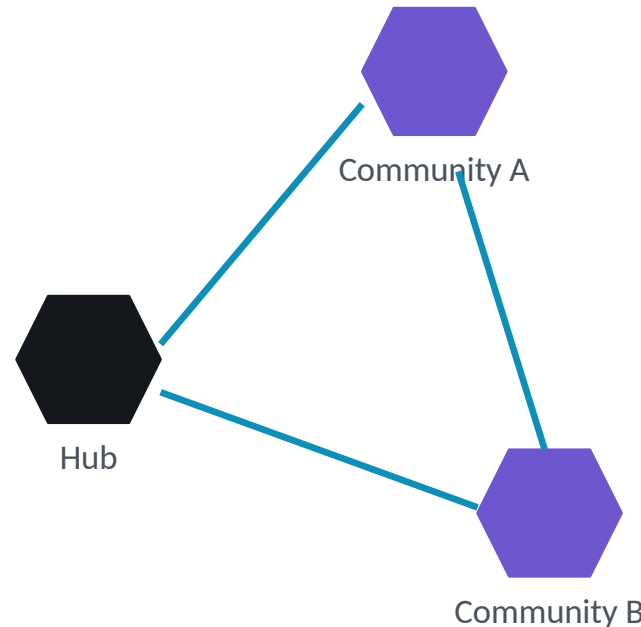
Every week, an optimiser plans the trips

Two separate trips



Two drives there and back

One shared trip



One loop: hub → A → B → hub

Google OR-Tools (CP-SAT) chooses which communities get a visit and which jobs are done, within each tradesperson's 40 hours, road closures and airstrips. **Trades going to the same place travel together: a further \$105 off every repair.**

\$76

saved per repair by shared trips, against the same plan without them

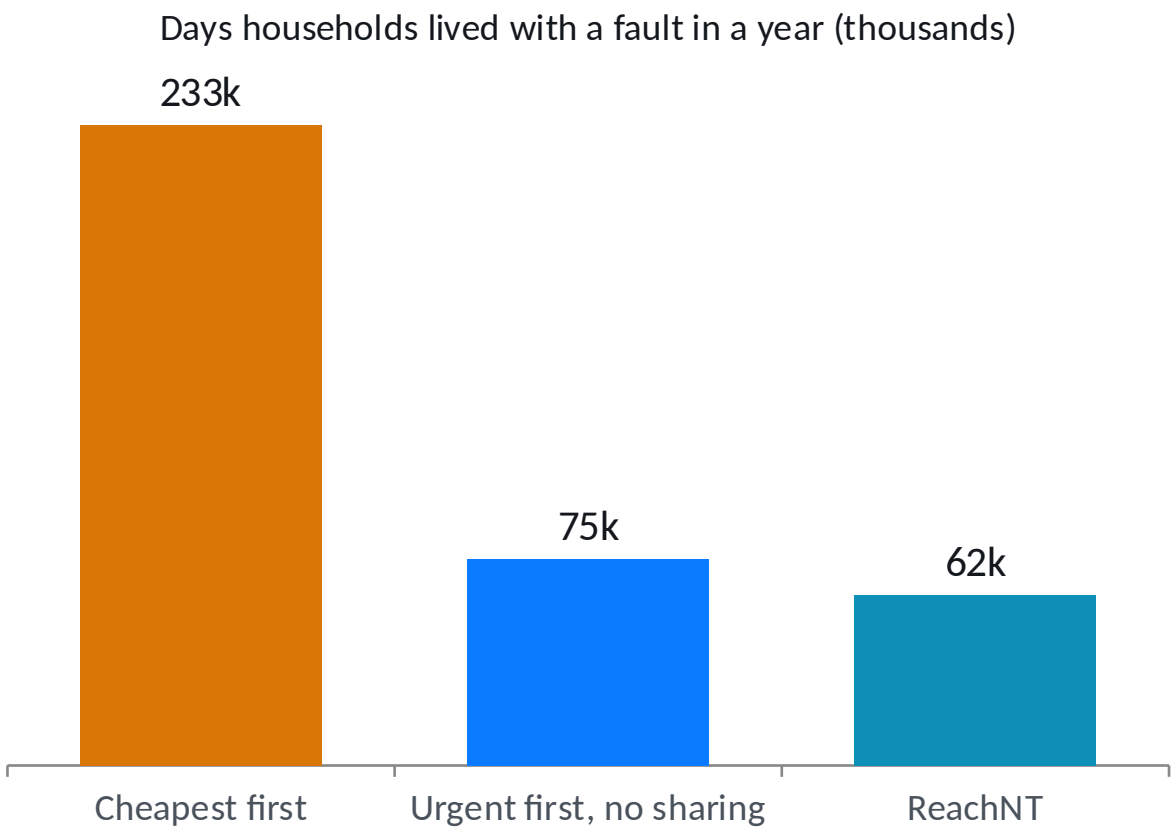
17%

fewer days households live with a fault, because the saving buys more visits

99.5%

of 26,018 weekly plans proved the best possible, in milliseconds

Fairness costs a little more, and saves months of waiting



	Cheapest first	ReachNT
Average cost per repair	\$561	\$821
9 in 10 urgent remote repairs fixed within	62 days	3 days
Days living with a fault	233k	62k

+47% cost, 74% fewer days with a fault

It held in all five random years we tested: 74%–77% fewer fault-days, for 45%–49% more per repair.

One simulated year on real NT geography: 37,508 synthetic repair requests, 70 communities, 5 trade hubs.

Floods: surge crews, not other communities' trades



Make every house safe within 48 hours

Maintenance and housing officers check each one; danger is made safe that day



One team, one trip

Plumber, electrician, carpenter and maintenance share the vehicle or the plane



Surge crews from the contractor panel

Local Aboriginal Business Enterprises first; a named person signs the request



Tell every household

Hotline, housing officer, free interpreters; every report tagged to the event

Live NT Road Report and Bureau of Meteorology readings prompt the coordinator: a road closure, heavy rain (“Declare a flood?”) or heat for Tier 1 homes. A person decides.

Modelled flood: Kalkarindji, Daguragu, Pigeon Hole

94 repairs, roads cut for 4 weeks (like January 2023). 9 in 10 urgent repairs fixed within:

In the flooded communities

14 days

usual crews

3 days

surge crews

Everywhere else in the hub (3 with no flood)

10 days

usual crews

3 days

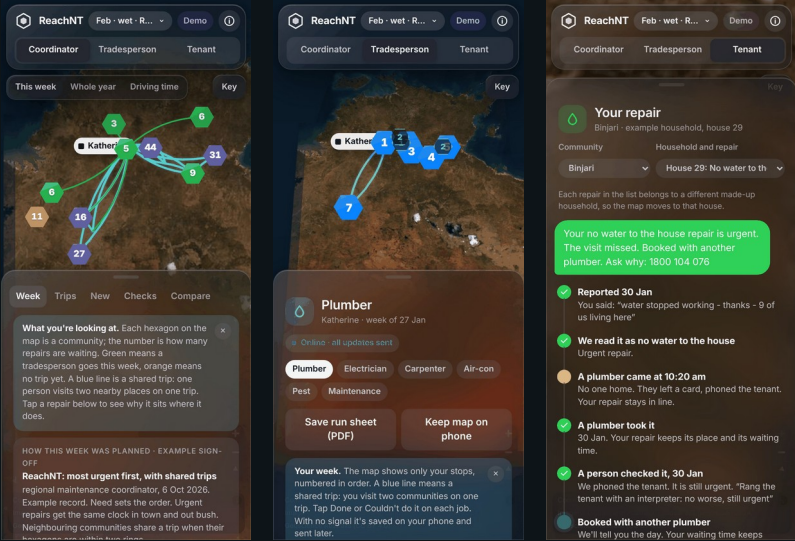
surge crews

Dangerous faults the reader recognised were made safe within a day (we assume a maintenance officer on hand).

Live now, on a real satellite map, and on any phone



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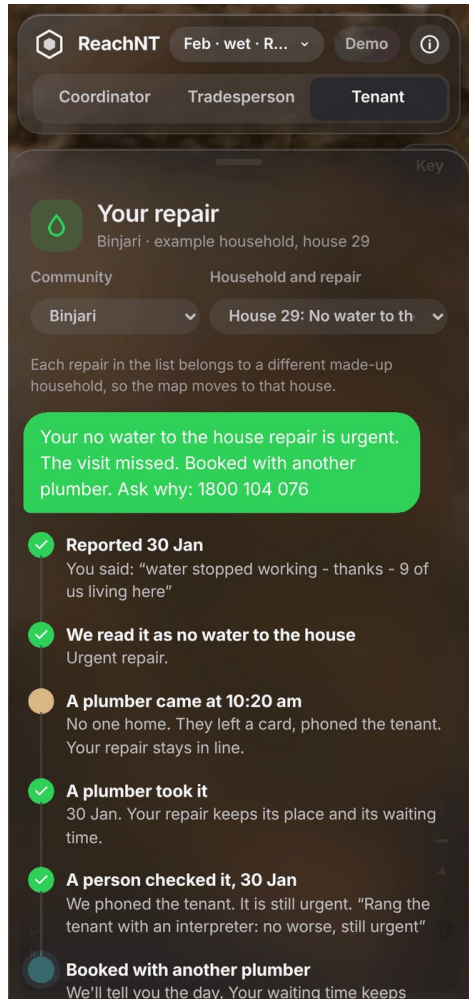
Coordinator

Tradesperson

Tenant

- Anyone the tenant tells can log a report, with the same questions
- Works without signal: updates are saved on the phone and sent later
- Plain words, WCAG 2.2 AA checked, PDFs, no confidence percentages

People stay in charge, and tenants can push back



A named person signs the trade-off

How much cost may count is a signed decision, kept in a tamper-evident record



A person can change urgency at any time

From a call, a visit, a photo or a review. Lowering danger needs someone who spoke to the tenant or saw it



A missed visit is never closed, and keeps its place

Time and what was tried are recorded; then the next trip, another crew, or anyone who can go sooner



No app needed: the housing officer acts for the tenant

Fixed, still broken, got worse or a review. A fix that didn't hold moves up the line



People check the AI every week

1 in 20 reports the computer read alone is re-read by a person



Personal details locked away

Names, phones and addresses in an encrypted vault; the house is just a hexagon

What we tested, and what we can't claim yet

Tested

- 148 automated checks on every code change: data, the reader, fair intake, the planner, the server
- Five random years of requests; the result held in all five
- Every community's position checked against satellite photos
- Accessibility: 0 WCAG 2.2 AA issues in every view
- Security: locked-down site, validated inputs, sign-in rules ready

Not yet

- The repair requests and households are synthetic: no public NT repair data exists
- The reader is tested in English only. Tenants speak Kriol, Aboriginal English and many languages
- No community has reviewed it yet. That comes first, not last
- Our test kit is ready the day real reports arrive, labelled with the Aboriginal Interpreter Service

What we recommend



Make the trade-off a signed decision

Publish the triage rules, ask every tenant the same questions however they report, and record who decides how much cost may count.



Pilot in one region, with communities

Start with the Katherine hub's 16 communities, co-designed with tenants, Aboriginal Housing NT, land councils and the Interpreter Service.



Test the AI on real words first

Score the reader on real reports in Kriol and Aboriginal English before it reads a single one on its own.

Where someone lives shouldn't decide how long they wait

ReachNT fixes 9 in 10 urgent remote repairs within 3 days instead of 62, for 47% more, and makes someone own that choice.

reachnt.vercel.app • github.com/harshrastogii/reachnt

Thank you. Questions welcome.

